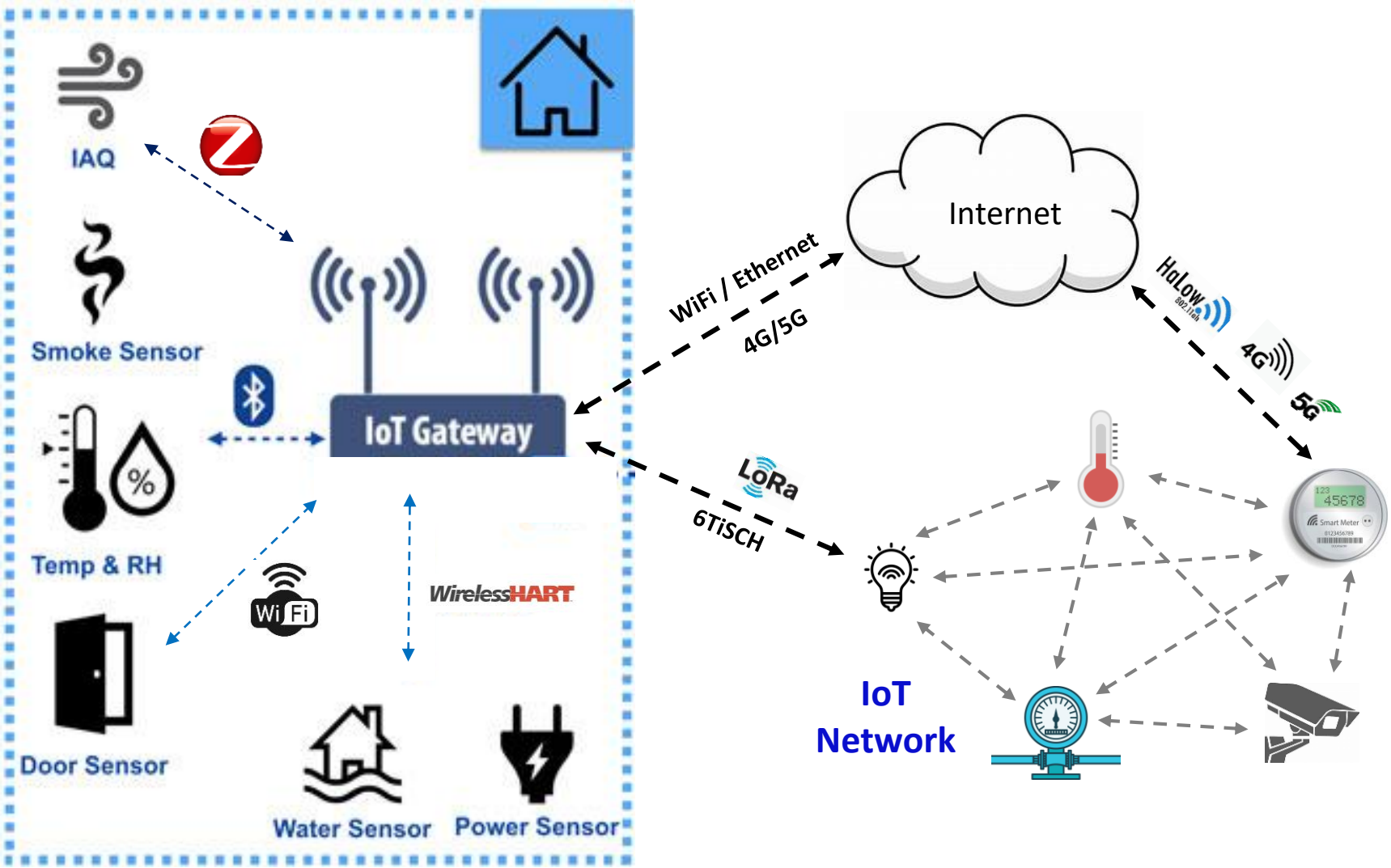
How IoT Works?

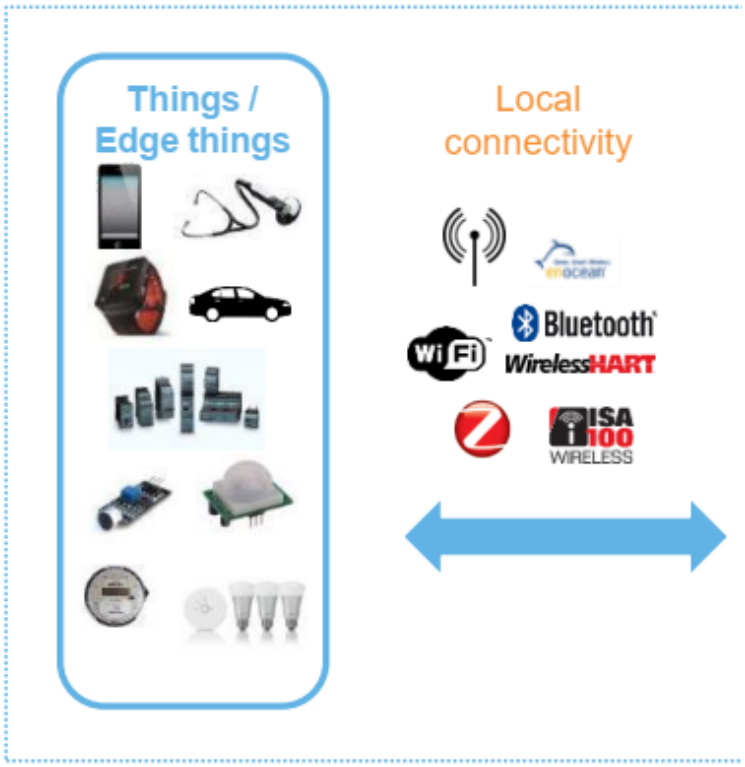
Smart Home IoT Application



Traditional Data Flow in IoT

Image Source: http://events17.linuxfoundation.org/sites/events/files/slides/Intelligence%20at%20the%20Edge.pdf

Traditional Data Flow in IoT



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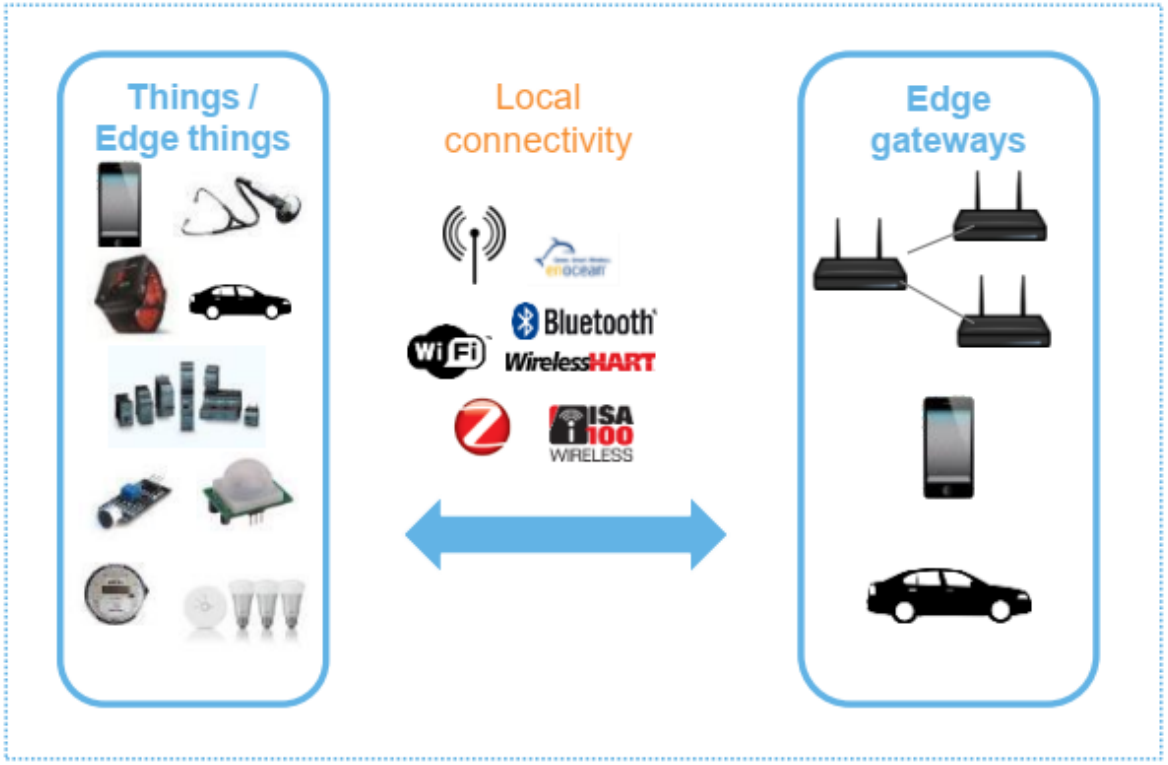

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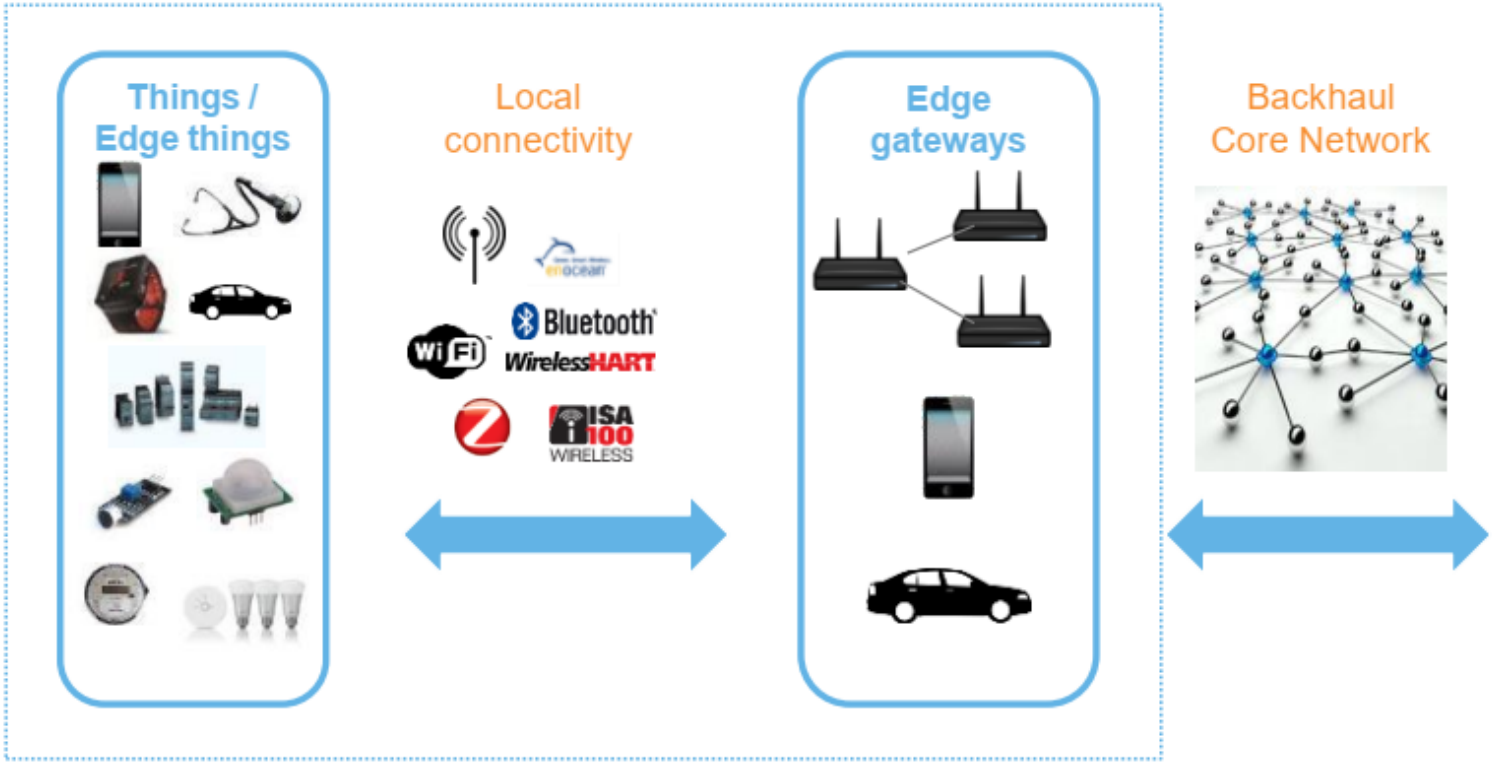

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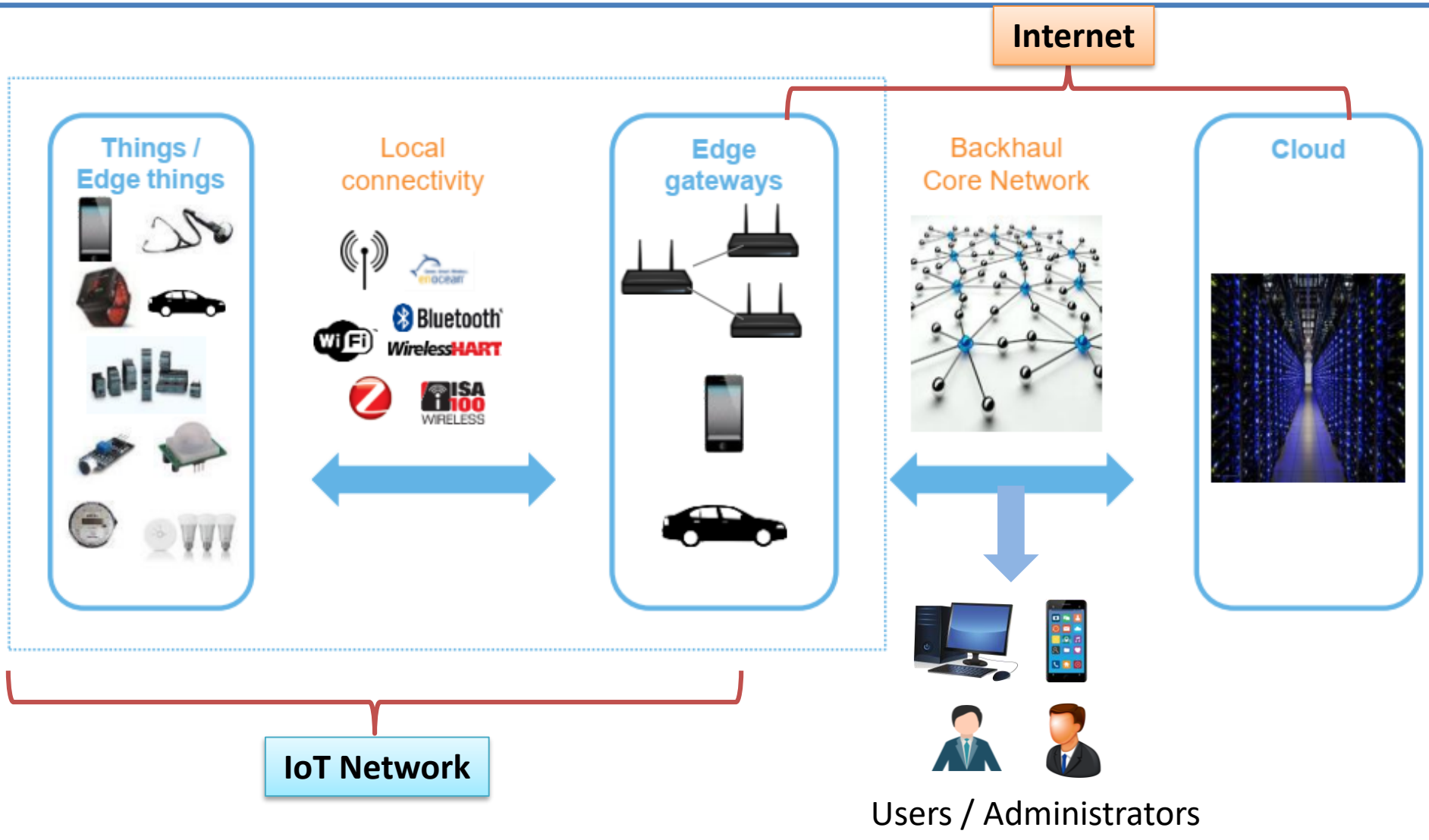

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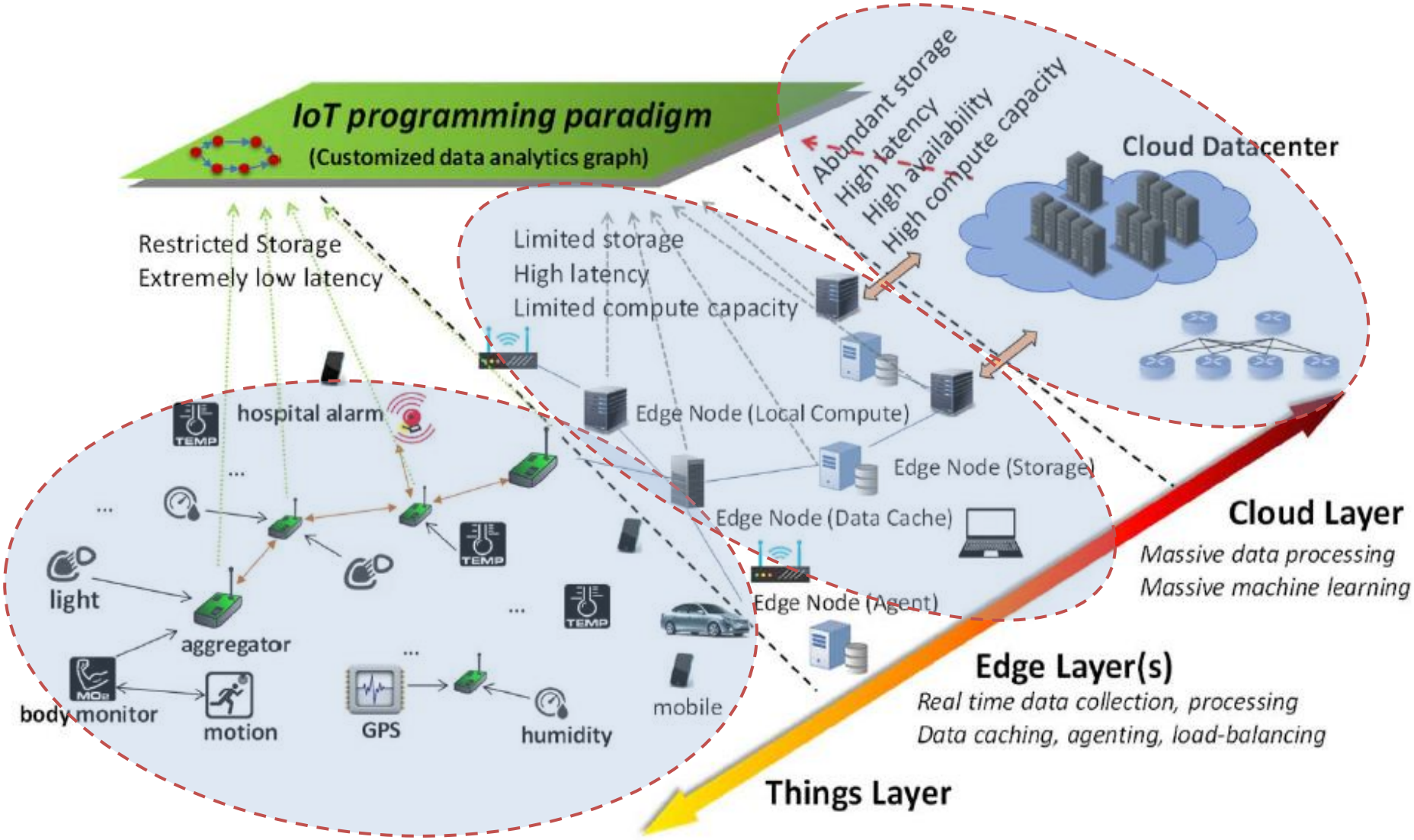
Data Processing Locations

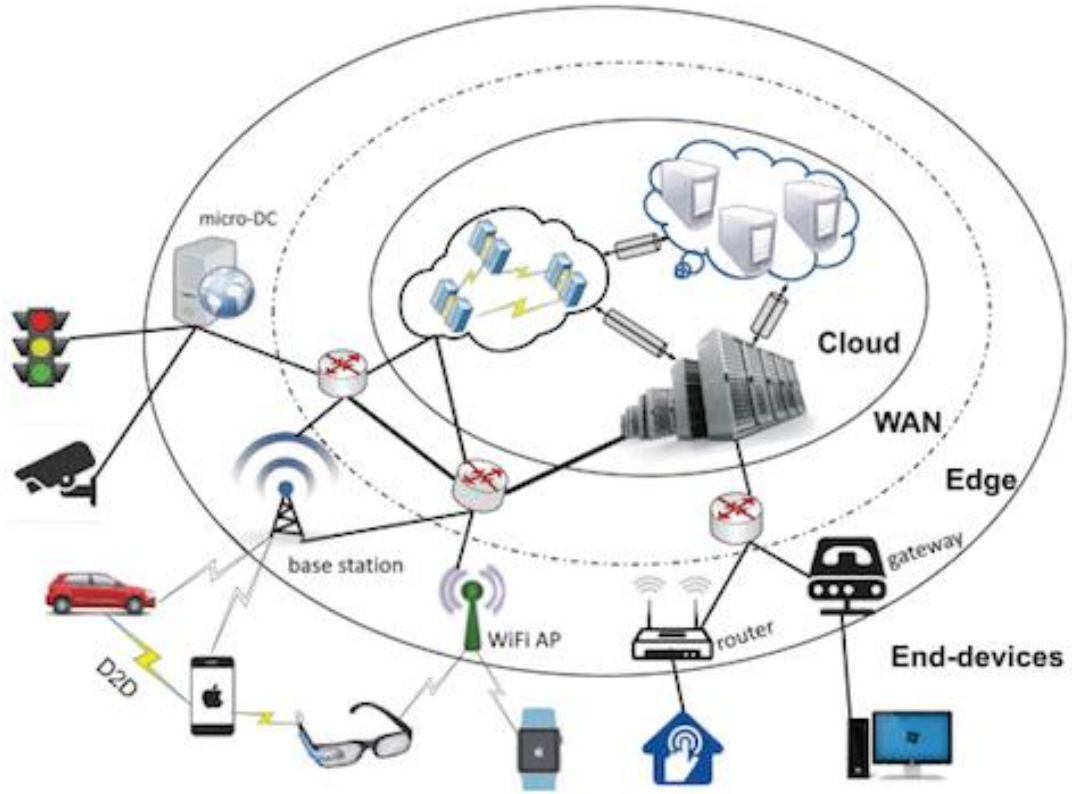


Source: Rajiv Ranjan *et. al.*, "Integrating the IoT and Data Science" *IEEE Cloud Computing, 2018*

Data Processing at the Edge

Edge Computing

- Host computation tasks close to the data sources and end users.



AI:

- Extracts insights that leads to better decisions & strategic move

Edge Intelligence:

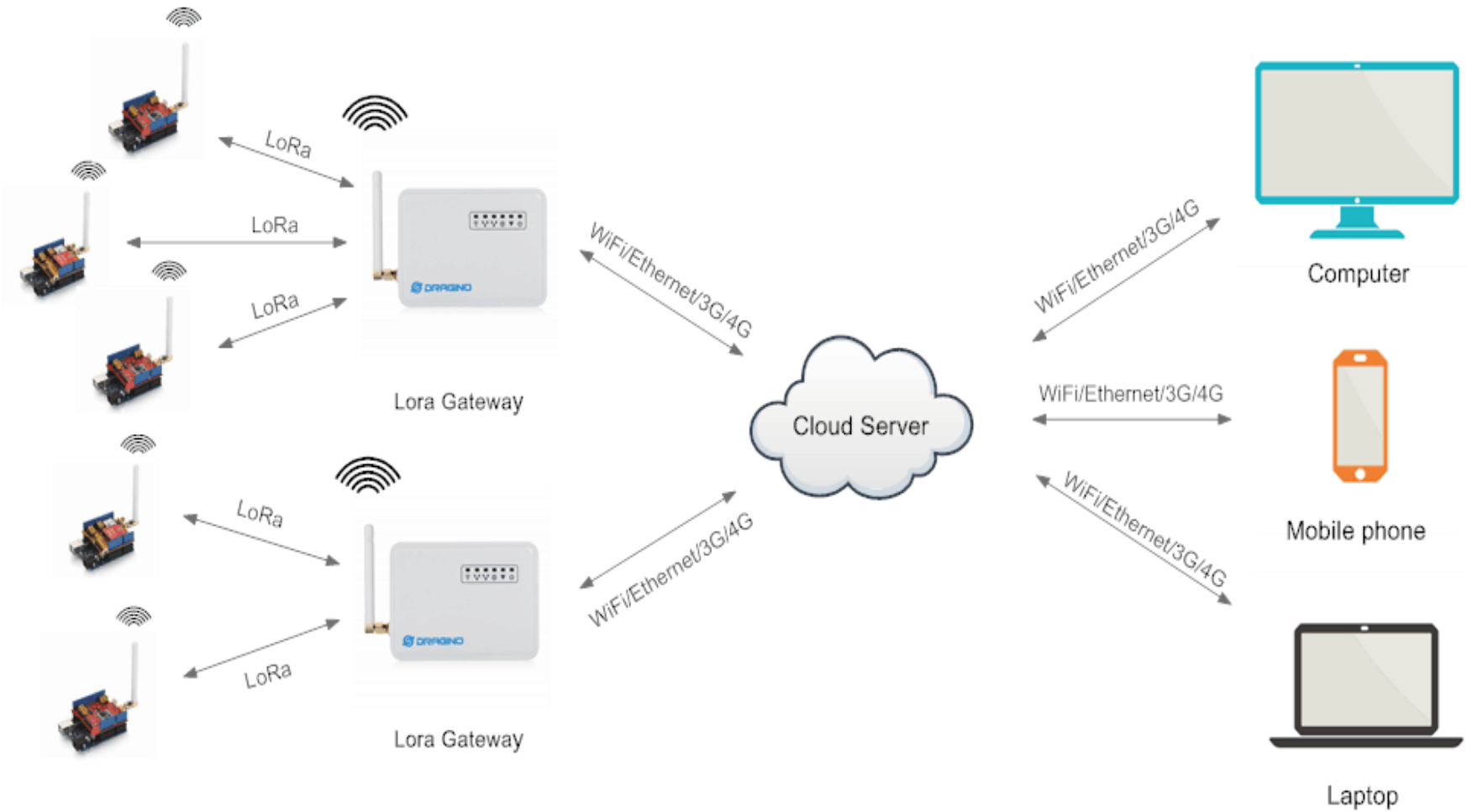
- Apply AI on data at network Edge

Image Source: Z. Zhou *et. al.* "Edge Intelligence: Paving the Last Mile of Artificial Intelligence With Edge Computing," in Proceedings of the IEEE, 107(8), pp. 1738-1762, Aug. 2019.

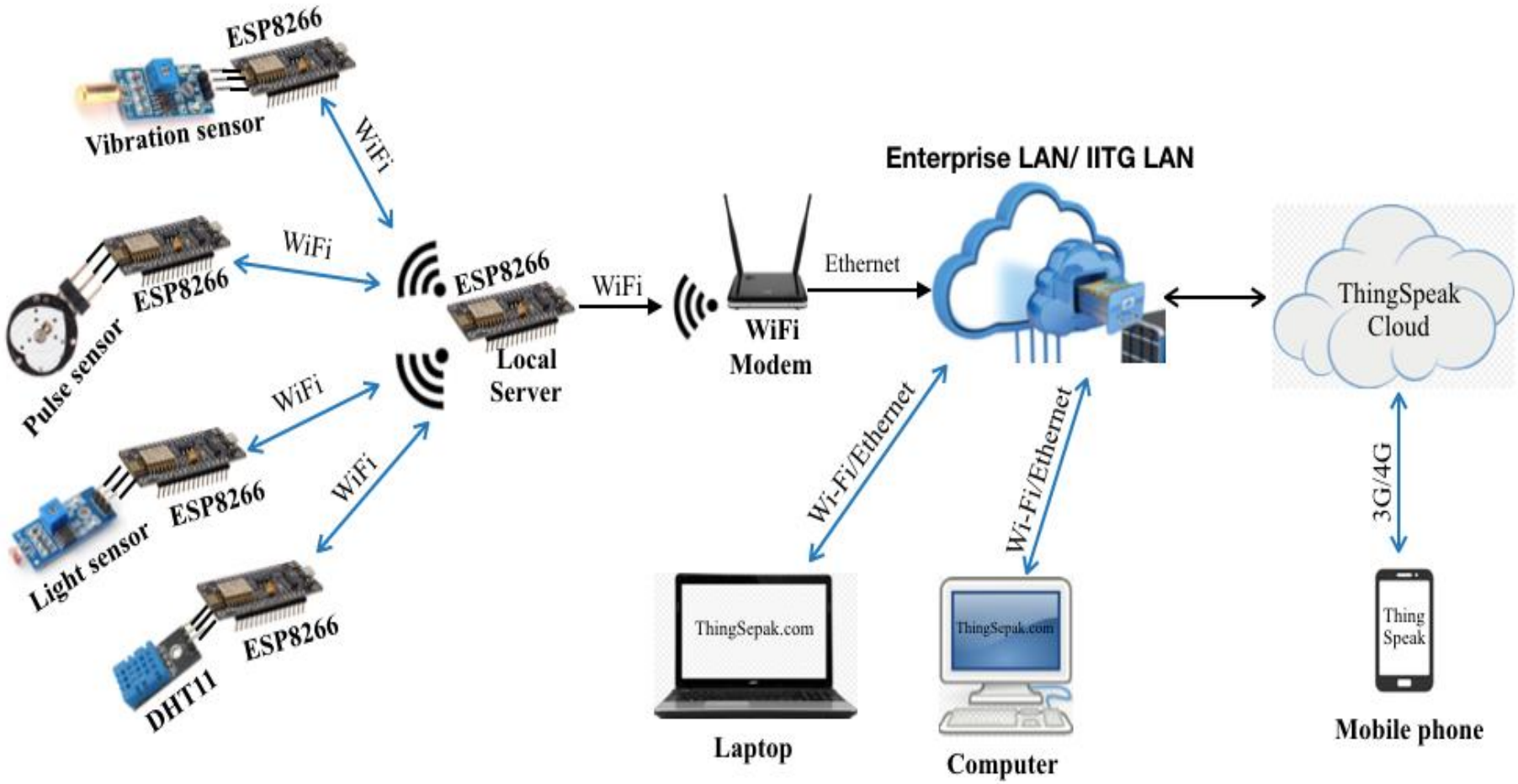
What is needed to know?

- **IoT Components**
  - Application developer needs to know what are the required components for designing an IoT application
- **IoT Architecture**
  - For any IoT application it is required to know the physical arrangement of IoT sensors, actuators, gateway, edge devices and Internet connectivity
- **IoT Network Access Technology**
  - Types of IoT Network Access Technology or Connectivity Technology used to connect the sensors/actuators with each other, and with the IoT gateway or Internet
- **IoT Data Processing / Analysis**
  - Sensed data needs to be stored and analyzed to take business decision or to get new business insight. Where it can be done and How?

DIY: Long-range Monitoring & Control



DIY: Smart Home Monitoring



Lessons Learned

- ✓ Layered View of an IoT Application
- ✓ Traditional Data Flow in IoT
- ✓ Data Processing Locations in IoT
- ✓ What is needed to know before developing IoT Applications

Thanks!